KEEGAN DEMPSEY, PE

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WORK EXPERIENCE

NON ALIGNED CLIMATE ACTION PROJECT (NACAP LLC), New York, NY, USA (November 2023 to Present)

Founder, Independent Research and Advisory

- Advised on technoeconomic model development for two early stage Australian hydrogen startups.
- Designed a decision framework for screening community scale waste to value pathways, integrating feedstock availability, process performance, policy constraints, socioeconomic drivers and local demand into an optimization ready model.
- Built a geospatial data pipeline converting public facility, wastewater, emissions, permitting and transport corridor records into a machine readable network of candidate sites, links and capacity bounds, paired with an interactive viewer for auditing inferred connections before optimization.
- Formulated an optimization tool with technoeconomic analysis at site level, made tractable by a route generation and pruning method that builds candidate pathways from the inferred infrastructure graph rather than a generic superstructure.
- Trained equipment level surrogate models to predict gasification and related conversion yields across heterogeneous feedstocks, reducing reliance on computationally intensive process simulations.

HYSATA, Port Kembla, NSW, AUS (February 2023 to August 2023)

Balance of Plant (BOP) & Systems Engineer / Technoeconomic Modeling Lead

- Modeled load following system models from first principles and experimental electrolyzer performance data to optimize BOP configuration for diverse renewable supply profiles.
- Devised BOP process simulation in Aspen HYSYS integrating dynamic electrolyzer degradation curves to inform equipment sizing, layout and capital cost estimation.
- Ideated, built and managed an integrated technoeconomic model, coupling scenario analysis and performance metrics to quantify the impact of technical and commercial decisions on levelized hydrogen cost.
- Proposed stack level design innovations to consolidate BOP functionalities, achieving cost & performance improvements.
- Aligned technical / commercial team workflows, bridging data and scope gaps to improve cross functional project execution.

ARUP, Melbourne, VIC, AUS & New York, NY, USA (September 2019 to January 2023)

Chemical Engineer, Energy, Transaction Advisory & Sustainability Groups

- Formulated technoeconomic modeling methods, electrolyzer performance models and plant process simulations for incentivizing, sizing and costing of green hydrogen projects worldwide.
- Advised governments and companies on hydrogen and synthetic fuel deployment strategies.
- Substantiated technical due diligence projects, verifying technical aspects of projects and technologies.
- Conducted and oversaw research to develop educational tools for internal and external stakeholders.

Mechanical Engineer, Buildings, Infrastructure & Sustainability Groups

- Developed energy models to reduce building energy loads and inform design.
- Led design and analysis of mechanical scope on critical infrastructure projects globally.

BURNS & MCDONNELL, New York, NY, USA & Houston, TX, USA (May 2015 to August 2019)

Assistant Mechanical Engineer, Aviation & Federal Group

- Collaborated on front end planning through detailed design for mechanical systems on the $4B Delta LaGuardia Airport Terminal Reconfiguration Project, coordinating piping, HVAC and fueling scope across a large multidisciplinary infrastructure team.
- Automated mechanical design models, improving workflow efficiency for the project team across disciplines.

Assistant Chemical Engineer, Oil, Gas & Chemicals Group

- Facilitated front end planning through detailed design of oil and gas processing plant projects.
- Created automation processes for P&ID models to include data in heat and material balances to generate equipment, line list and pipe sizing calculations resulting in improved workflow efficiency across groups.

EDUCATION

CARNEGIE MELLON UNIVERSITY, Pittsburgh, PA, USA (December 2026, expected)

Master of Science, Chemical Engineering

Thesis: Context Driven Process Synthesis for Waste Valorization, advised by Ignacio Grossmann

TEXAS TECH UNIVERSITY, Lubbock, TX, USA (May 2015)

Bachelor of Science, Chemical Engineering (Minor: Bioengineering, Chemistry)

SKILLS

Software: Aspen (HYSYS, PLUS, etc.), IES, COMSOL, Smart Plant P&ID, Revit, UniSim, Excel, LLM Agents (Codex, Claude Code)
Programming: Python (Pyomo, IDAES, causal-learn, scikit-learn, JAX, etc.), GAMS, Matlab.

CONFERENCE PRESENTATIONS & PUBLICATIONS

- UKPACT. *eThekwini Regional Hydrogen Strategy.* eThekwini Municipality Energy Office, 2023. Link.
- Dempsey, K. & Gillespie, J. *" Can Synthetic Fuels Help Reduce Emissions in Australia's Rail Sector While Paving the Way to a Carbon Free Future?".* Presentation at *AUSRail,* 6 Dec. 2022, Brisbane, QLD. Conference Paper. Link.
- Department of Climate Change, Energy, the Environment and Water, and Australian Hydrogen Council. *Water for Hydrogen: Technical Paper.* Reference: 286739-00-RPT-001. 14 Nov. 2022, Link.
- Dempsey, K. *Cleaner, Alternative Fuels for Transport.* Presentation at *Our Pathway Through Net Zero: Reduce, Restore, Remove: A Call to Action,* 30 Aug. 2022, Melbourne, VIC. Video. Link.
- World Bank. *Opportunities and Barriers for the Deployment of Green Hydrogen in Chile's Markets: Small and Medium Grids: Task 3: Isolated Grid Stud*y. Washington, D.C.: World Bank Group, 01 Aug. 2022. Link.